

T3 matched BSM $\Delta\mathcal{L}_{\text{EFT}}$ terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from matched BSM EFT terms. Empty cells denote no matched term with that field combination.

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
									$+C_{H3}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{\text{lambdaH1Inv1}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{5\text{lambdaH1Inv2}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon}$ $+\frac{\text{lambdaH2Inv1}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{5\text{lambdaH2Inv2}^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon}$ $+\frac{\text{lambdaH1Inv1lambdaH1Inv2}2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{\text{lambdaH2Inv1lambdaH2Inv2}2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{1}{2}\text{lambdaH1Inv1}^2h\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{5}{6}\text{lambdaH1Inv2}^2h\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{\text{lambdaH1Inv1lambdaH1Inv2}h\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}}$ $+\frac{1}{2}\text{lambdaH2Inv1}^2h\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{5}{6}\text{lambdaH2Inv2}^2h\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{\text{lambdaH2Inv1lambdaH2Inv2}h\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}}$ $+\frac{h\overline{\lambda_{\tau 3}}H_{a_1}H_{a_2}\lambda_{\tau 3}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{h\overline{\lambda_{\tau 3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau 3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $+\frac{h\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)\overline{\lambda_{\tau 3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau 3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $-\frac{h\overline{\lambda_{\tau 3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\tau 3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $-\frac{h\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)\overline{\lambda_{\tau 3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\tau 3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$	$-\frac{h\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{12\epsilon}$ $-\frac{1}{24}h\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$ $-\frac{1}{24}h\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{h\log\left(\frac{M_F^2}{M_\ell^2}\right)\overline{y_{\tau 1}\tau_1y_{\tau 2}\tau_2}\ell_{a_3\tau_2}^\tau C_P\ell_{a_2\tau_1}H_{a_1}M_{S_1}^2\lambda_{\tau 1}\ell_{a_1\tau_2}\ell_{a_3\tau_4}M_F^2}{(M_\ell^2-M_{S_1}^2)(M_\ell^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{h\log\left(\frac{M_F^2}{M_\ell^2}\right)\overline{y_{\tau 1}\tau_1y_{\tau 2}\tau_2}\ell_{a_3\tau_2}^\tau C_P\ell_{a_2\tau_1}H_{a_1}M_{S_1}^2\lambda_{\tau 1}\ell_{a_1\tau_2}\ell_{a_3\tau_4}M_F^2}{(M_\ell^2-M_{S_1}^2)(M_\ell^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{h\log\left(\frac{M_F^2}{M_\ell^2}\right)\overline{y_{\tau 1}\ell_{a_2\tau_1}}C_P\overline{y_{a_4\tau_2}}M_F^2y_{\tau_1\tau_1}y_{\tau 2}\tau_2\tau_1\ell_{a_1\tau_2}\ell_{a_3\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{(M_\ell^2-M_{S_1}^2)(M_\ell^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{h\log\left(\frac{M_F^2}{M_\ell^2}\right)\overline{\lambda_{\tau 1}\ell_{a_2\tau_1}}C_P\overline{y_{a_4\tau_2}}M_F^2y_{\tau_1\tau_1}y_{\tau 2}\tau_2\tau_1\ell_{a_1\tau_2}\ell_{a_3\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{(M_\ell^2-M_{S_1}^2)(M_\ell^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{h\log\left(\frac{M_F^2}{M_\ell^2}\right)\overline{y_{\tau 1}\tau_1y_{\tau 2}\tau_2}\ell_{a_3\tau_2}^\tau C_P\ell_{a_2\tau_1}H_{a_1}M_{S_1}^2M_\ell^2\lambda_{\tau 1}\ell_{a_1\tau_2}\ell_{a_3\tau_4}M_F}{(M_\ell^2-M_{S_1}^2)(M_\ell^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{h\log\left(\frac{M_F^2}{M_\ell^2}\right)\overline{y_{\tau 1}\tau_1y_{\tau 2}\tau_2}\ell_{a_3\tau_2}^\tau C_P\ell_{a_2\tau_1}H_{a_1}M_{S_1}^2M_\ell^2\lambda_{\tau 1}\ell_{a_1\tau_2}\ell_{a_3\tau_4}M_F}{(M_\ell^2-M_{S_1}^2)(M_\ell^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{h\log\left(\frac{M_F^2}{M_\ell^2}\right)\overline{\lambda_{\tau 1}\ell_{a_2\tau_1}}C_P\overline{y_{a_4\tau_2}}M_\ell^2M_{S_1}^2y_{\tau_1\tau_1}y_{\tau 2}\tau_2\tau_1\ell_{a_1\tau_2}\ell_{a_3\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_\ell^2-M_{S_1}^2)(M_\ell^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{h\log\left(\frac{M_F^2}{M_\ell^2}\right)\overline{\lambda_{\tau 1}\ell_{a_2\tau_1}}C_P\overline{y_{a_4\tau_2}}M_\ell^2M_{S_1}^2y_{\tau_1\tau_1}y_{\tau 2}\tau_2\tau_1\ell_{a_1\tau_2}\ell_{a_3\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_\ell^2-M_{S_1}^2)(M_\ell^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	e
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